PROJECT REPORT — ATG TECHNOLOGIES

LaunchPad. *Job Portal*

A focused platform for discovering internships and early-career opportunities, built with the MERN stack.

DEVELOPED BY

Keyur Patil

ORGANISATION

ATG Technologies

TECHNOLOGY STACK

MERN (MongoDB, Express, React, Node.js)

DEPLOYMENT

Render (Frontend + Backend) · MongoDB Atlas

REPOSITORY

github.com/keyurpatil2006-sudo/launchpad-job-portal

REPORT DATE

November 2026

01 PROJECT OVERVIEW

What is *LaunchPad?*

LaunchPad is a full-stack internship and job opportunity platform built to connect early-career candidates with meaningful roles. The platform presents a curated collection of internships and full-time positions across Engineering, Design, Marketing, Data & Analytics, and Content domains.

Candidates can search, filter, and explore role detail pages with editorial-style content — including company stories, 30-day onboarding timelines, learning outcomes, and culture values — before submitting a guided multi-step application. Employers and administrators access a protected workspace where they can create, edit, and delete listings, and review all incoming applications.

Project Objectives



Opportunity Discovery

Provide a clean, searchable, and filterable listing of internships and early-career roles that candidates can explore with minimal friction.



Streamlined Applications

Guide candidates through a 3-step application flow with resume upload, cover notes, and submission confirmation.



Employer Workspace

Give authenticated administrators a secure dashboard to manage listings and monitor incoming applications in real time.



Mobile-First Responsive UI

Ensure the entire platform is fully usable on smartphones and tablets with an adaptive, touch-friendly interface.

Live Project URLs

The project is deployed and publicly accessible:

Frontend: `https://launchpad-job-portal-1.onrender.com`

Backend API: `https://launchpad-job-portal.onrender.com/api`

02 FEATURES & FUNCTIONALITY

What the *platform does.*

For Candidates

1 **Search & Filter Opportunities**

Search by role title, company name, or keyword. Filter by domain (Engineering, Design, Marketing, Data & Analytics, Content) and job type (Internship, Full-time, Part-time). Results update instantly.

2 **Featured Opportunity Spotlight**

A highlighted "Opportunity of the Week" card with editorial-style design, company logo, metadata, and a direct call-to-action for high-intent candidates.

3 **Editorial Role Detail Pages**

Each listing has a full-page editorial profile including: role description, company story, learning outcomes, 30-day onboarding roadmap, culture values, perks, and a sticky application sidebar.

4 **Multi-Step Application Form**

A guided 3-step application: (1) Profile — name & email, (2) Resume — upload PDF/DOC or paste a link, add cover note, (3) Review & submit. Applications are submitted via authenticated API and stored in MongoDB.

5 **User Accounts**

Candidates can register and log in. JWT tokens are stored in localStorage and sent with API requests to authenticate submissions.

For Employers & Administrators

A **Protected Admin Workspace**

Admins sign in with configured credentials. The admin route is protected server-side by JWT middleware; unauthenticated requests are rejected with a 401 response.

B **Opportunity Management (CRUD)**

Create new opportunities via a modal form. Edit existing listings inline. Delete listings with confirmation. All changes persist to MongoDB instantly.

C **Application Inbox**

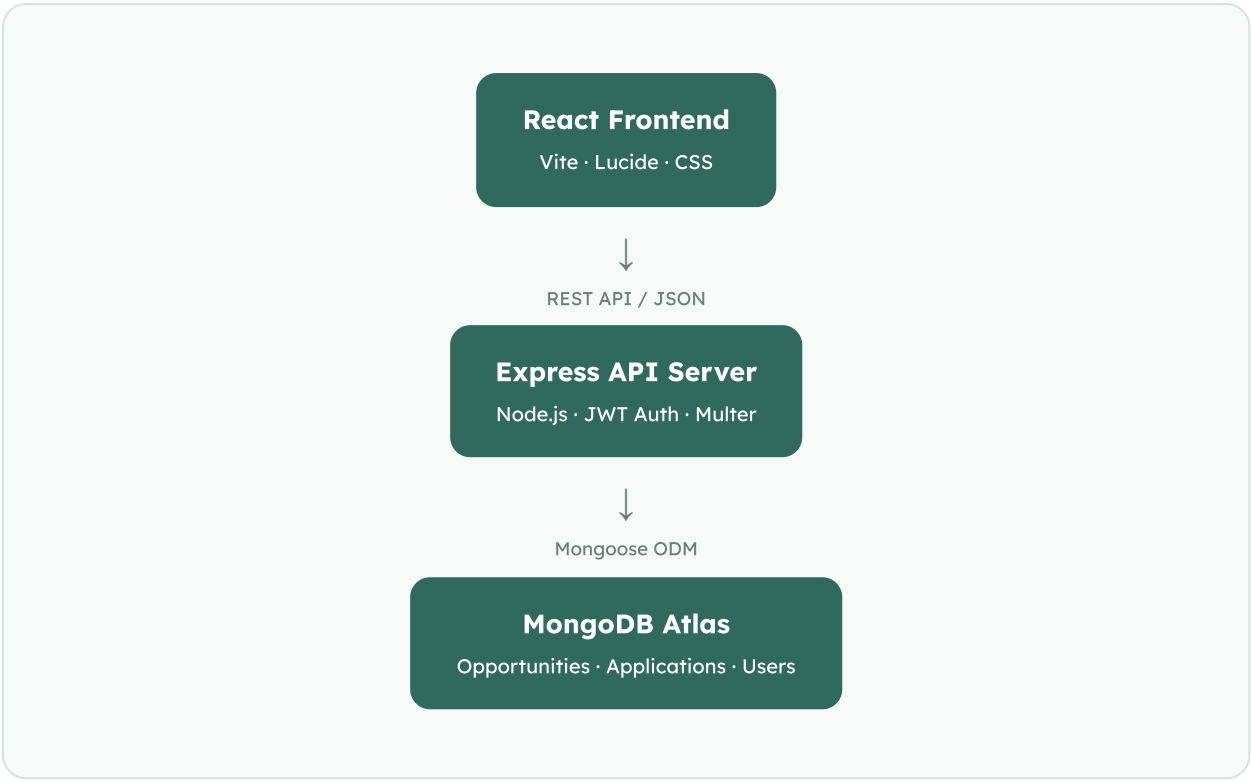
View all submitted applications in a table: applicant name, email, the opportunity applied to, application status, and submission date.

03 TECHNOLOGY STACK & ARCHITECTURE

How it *is built.*

LAYER	TECHNOLOGY	PURPOSE
Frontend Framework	React 18 + Vite	Component-based SPA with fast HMR dev server and optimised production builds
UI Icons	Lucide React	Consistent, lightweight icon set used throughout the interface
Styling	Custom CSS (CSS Variables)	Responsive layouts using media queries; no CSS framework dependency
Backend Runtime	Node.js 20 + Express	REST API server handling auth, opportunities, and application routes
Database	MongoDB + Mongoose	Document store for opportunities, applications, and users
Authentication	JWT + bcryptjs	Stateless token auth; passwords hashed with bcrypt before storage
File Upload	Multer	Handles multipart form data for resume file uploads
Email (optional)	Nodemailer	Application notification emails via SMTP
Dev Tooling	Nodemon + Concurrently	Auto-restart server; run frontend and backend in parallel
Hosting	Render	Frontend static site + backend web service on Render
Database Hosting	MongoDB Atlas	Cloud-managed MongoDB cluster
Source Control	Git + GitHub	Version control; Render auto-deploys on push to main branch

System Architecture



04 PROJECT STRUCTURE & API REFERENCE

Code & *endpoints.*

Folder Structure

```
launchpad-job-portal/
├── client/  # React + Vite frontend
│   ├── src/
│   │   ├── main.jsx  # All React components
│   │   └── styles.css  # Responsive styles
│   └── index.html
├── server/  # Node.js + Express backend
│   ├── src/
│   │   ├── index.js  # Express app + all routes
│   │   └── seed.js  # Database seeder
│   ├── models/
│   │   ├── Opportunity.js
│   │   ├── Application.js
│   │   └── User.js
│   └── .env.example
├── package.json  # Root scripts
└── README.md
```

API Endpoints

AUTHENTICATION — /API/AUTH

Method	Endpoint	Auth	Description
POST	/auth/register	None	Register a new user account
POST	/auth/login	None	Login — returns a signed JWT token

OPPORTUNITIES — /API/OPPORTUNITIES

Method	Endpoint	Auth	Description
GET	/opportunities	None	List all opportunities
POST	/opportunities	Admin JWT	Create a new opportunity listing

Method	Endpoint	Auth	Description
PUT	/opportunities/:id	Admin JWT	Update an existing opportunity
DEL	/opportunities/:id	Admin JWT	Delete an opportunity

APPLICATIONS — /API/APPLICATIONS

Method	Endpoint	Auth	Description
POST	/applications	User JWT	Submit an application (multipart/form-data)
GET	/applications	Admin JWT	List all submitted applications

05 UI SCREENSHOTS — CANDIDATE EXPERIENCE

The *interface.*

Home Page — Opportunity Discovery

SCREENSHOT — HOME PAGE

Shows the hero search bar, type filter tabs, featured spotlight card, and opportunity grid

Featured Opportunity Spotlight

SCREENSHOT — SPOTLIGHT SECTION

Featured "Opportunity of the Week" card with editorial design and animated elements

05 UI SCREENSHOTS — ROLE DETAILS & APPLICATION

Role Detail Page (Editorial Layout)

SCREENSHOT — ROLE DETAIL PAGE

Editorial role profile with company story, 30-day roadmap, learning grid, culture values, and sticky apply sidebar

Multi-Step Application Form

SCREENSHOT — APPLICATION FORM (3-STEP)

Step 1: Profile | Step 2: Resume & cover note | Step 3: Review & submit

05 UI SCREENSHOTS — AUTH & ADMIN

User Login & Registration



SCREENSHOT — USER LOGIN / REGISTER

JWT-based login form for candidates with toggle between Sign In and Create Account

Admin Workspace — Opportunity Management

SCREENSHOT — ADMIN DASHBOARD

Protected employer workspace with opportunity table, Add / Edit / Delete controls, and stats summary

Admin Application Inbox

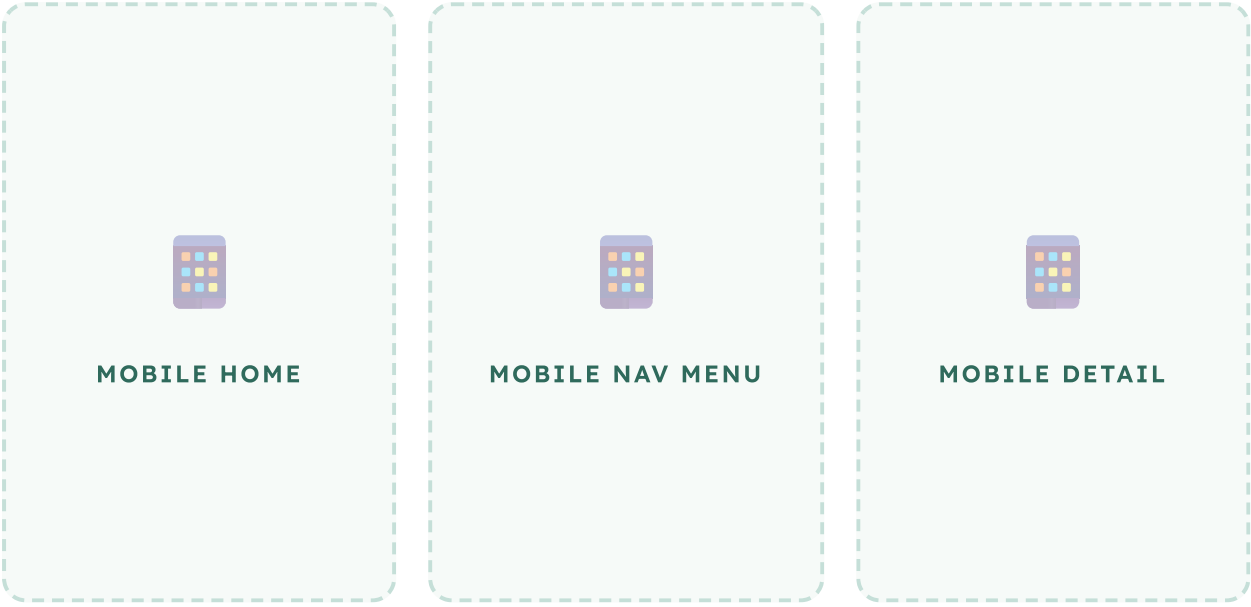


SCREENSHOT — APPLICATION INBOX

Table of submitted applications with applicant name, email, opportunity, status, and date

05 UI SCREENSHOTS — MOBILE RESPONSIVE UI

The platform is fully responsive across smartphones and tablets. A hamburger navigation drawer, single-column layouts, and fluid input controls ensure the complete candidate and admin experience works on mobile devices.



💡 **Mobile features added:** Viewport meta tag, hamburger drawer navigation with slide-down animation, single-column card grid, full-width search bar, responsive hero typography, touch-friendly buttons, and mobile-optimised form layouts.

06 SETUP & DEPLOYMENT

Running the *project.*

Local Development Setup

1. **Clone the repository**
   git clone https://github.com/keyurpatil2006-sudo/launchpad-job-portal.git

2. **Configure environment variables**
   Copy server/.env.example to server/.env and fill in your MongoDB URI, JWT secret, and admin credentials.

3. **Install all dependencies**
   npm run install:all — installs packages for root, client, and server in one command.

4. **Seed the database (optional)**
   npm run seed — loads initial opportunity data into MongoDB.

5. **Start development servers**
   npm run dev — runs the frontend on port 5173 and the API on port 5000 concurrently.

Environment Variables

```
# server/.env
PORT=5000
MONGO_URI=mongodb://127.0.0.1:27017/launchpad
JWT_SECRET=replace-with-a-long-random-secret
ADMIN_EMAIL=admin@example.com
ADMIN_PASSWORD=change-me-now

# client (Render env var)
VITE_API_URL=https://launchpad-job-portal.onrender.com/api
```

⚠️ **Security note:** Never commit your .env file or real credentials to version control. Add server/.env to .gitignore.

Render Deployment

BACKEND — WEB SERVICE

ROOT DIRECTORY	`server`
BUILD COMMAND	`npm install`
START COMMAND	`npm start`
ENVIRONMENT VARS	MONGO_URI, JWT_SECRET, ADMIN_EMAIL, ADMIN_PASSWORD

FRONTEND — STATIC SITE

ROOT DIRECTORY	`client`
BUILD COMMAND	`npm install && npm run build`
PUBLISH DIRECTORY	`dist`
ENVIRONMENT VARS	VITE_API_URL

07 FUTURE IMPROVEMENTS

What comes *next.*

- [ ] Pagination and infinite scroll for the opportunity listing grid
- [ ] Richer search filters: city, work mode (remote / hybrid / on-site), salary range
- [ ] Candidate profiles with saved and bookmarked opportunities
- [ ] Persistent file storage for uploaded resumes using AWS S3 or Cloudinary
- [ ] Email confirmation sent to candidates on successful application submission
- [ ] Automated integration tests in CI pipeline (GitHub Actions)
- [ ] Admin analytics dashboard — applications per listing, conversion rates
- [ ] Application status tracking for candidates (Received → Under Review → Decision)

08 CONCLUSION

What was *achieved.*

LaunchPad successfully demonstrates a complete, production-deployed MERN stack application covering the full development lifecycle — from local development and database modelling through REST API design, JWT authentication, file handling, responsive UI development, and cloud deployment.

The project prioritises user experience with an editorial design language, smooth animations, and a fully mobile-responsive interface. The employer admin workspace provides real-world CRUD functionality protected by token-based authentication. All changes are version-controlled on GitHub and auto-deployed to Render on every push to the main branch.

This internship project at ATG Technologies provided hands-on experience with every layer of modern web application development — reinforcing skills in React, Node.js, MongoDB, REST API design, responsive CSS, cloud deployment, and collaborative version control.

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Intern — ATG Technologies · November 2026

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